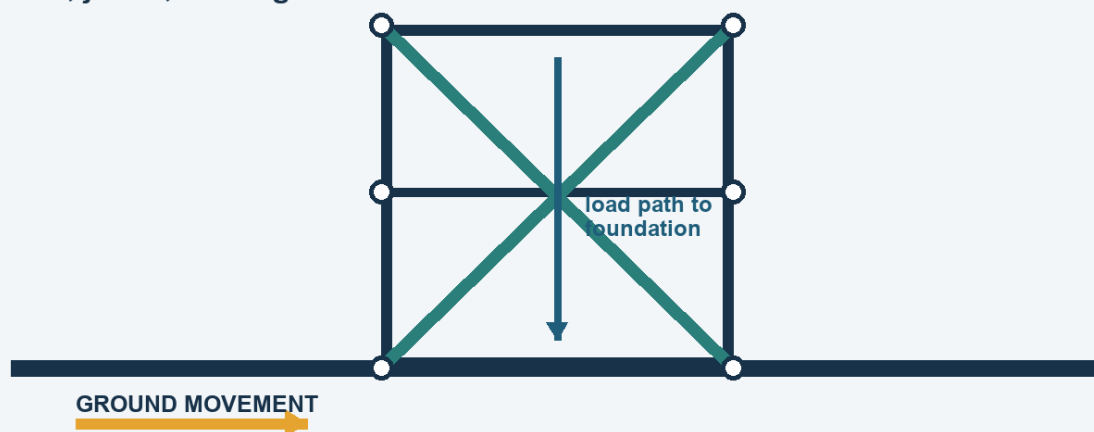


QUAKELAB INVESTIGATION

Earthquake Engineering

Structural basics through experimentation — then an open design challenge

Earthquake forces need a continuous route through floors, joints, bracing and foundations.



Student	Class	Teacher	Date

Driving question

How can evidence about forces, shape, materials, mass and vibration help us design safer buildings?

Year 9 Science & Engineering • 5–7 lessons • Individual task

START HERE

Your engineering mission

You will build knowledge in four controlled investigations before applying it to your own solution.

Learning intentions

- Explain how earthquake shaking creates forces in a structure.
- Investigate how bracing, materials, mass position and frequency affect structural response.
- Collect sensor data, identify patterns, and make evidence-based claims.
- Research earthquake-resistant design principles and judge the credibility of sources.
- Design, test and improve a solution within realistic constraints.

Success criteria

I can...	Evidence to include
conduct a fair test	one independent variable; other settings controlled
use measurements	units, sensor position, maximum response and cost
explain cause and effect	Claim–Evidence–Reasoning (CER), not just ‘it worked’
improve a design	labelled versions and a justified change
use research responsibly	credible sources, notes in my own words, links recorded

Important model limitation

QuakeLab is a simplified classroom model. A ‘survived’ result is not proof that a real building is safe or code-compliant. Real engineers use site data, standards, detailed analysis, testing and professional judgement.

Simulation survival checks

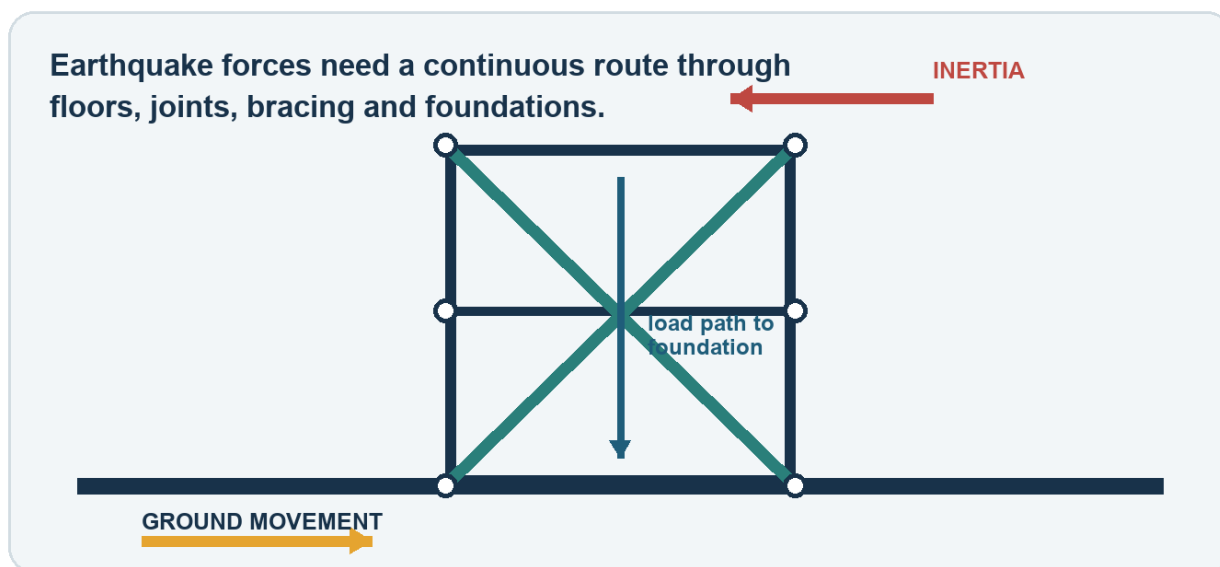
The app reports ‘survived’ only when all four simplified limits are met:

Roof sway	Storey drift	Failed members	Height retained
≤ 160 mm	≤ 80 mm	≤ 25%	≥ 65%

RESEARCH PRIMER

What happens during an earthquake?

Read, annotate the image, then answer using scientific language.



Ground motion travels as seismic waves. Buildings resist rapid movement because of inertia, creating changing horizontal forces. Damage depends on the motion's amplitude, frequency and duration, plus distance, depth, local geology, foundations, structural form and construction quality.

Australian context

Australia is not on a major plate boundary, but damaging earthquakes do occur. The 1989 Newcastle earthquake caused deaths and major damage. Earthquake hazard information is used in mitigation and building-code decisions.

R1. Distinguish earthquake magnitude from shaking intensity at a particular place.

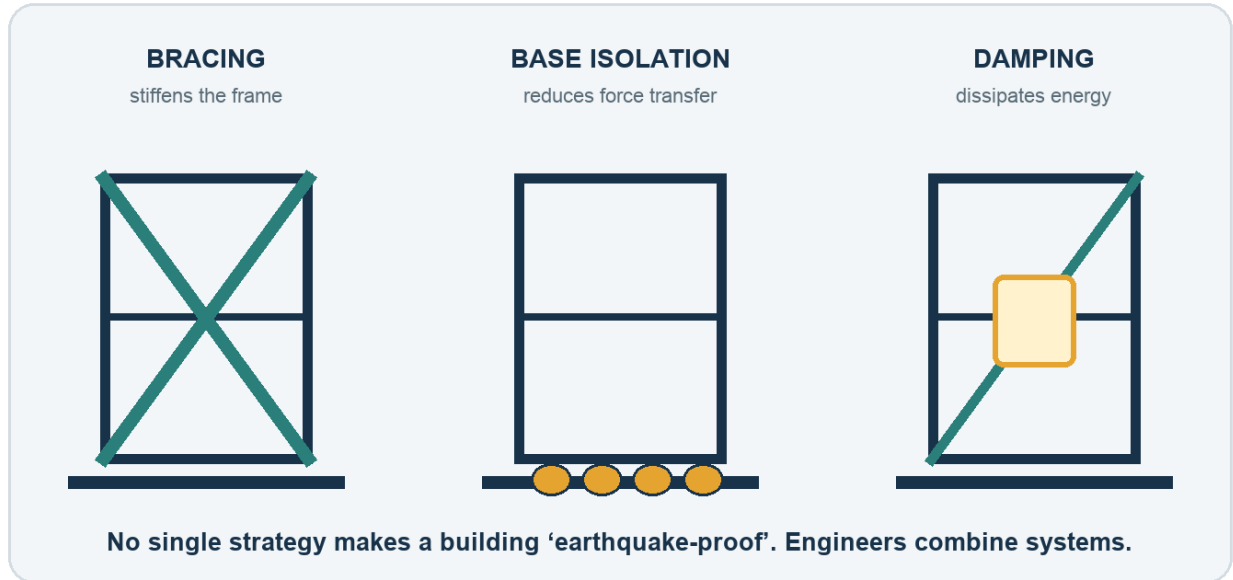
R2. Use the diagram to explain why a continuous load path from roof to foundation matters.

R3. Name two site factors and two building factors that could change the amount of damage.

STRUCTURAL BASICS

Five ideas engineers combine

A safer design balances performance, cost, materials, use and constructability.



Principle	Meaning	Possible strategy
Load path	Forces travel through connected parts to the ground.	Strong joints; tied floors/walls; secure foundations
Strength + stiffness	Strength resists failure; stiffness limits shape change and drift.	Braced frames; shear walls; adequate members
Ductility	A system deforms and carries load instead of failing suddenly.	Detailed steel/reinforcement; ductile connections
Energy control	Reduce transferred energy or dissipate it.	Base isolation; dampers; material damping
Regularity + mass	Balanced load paths and sensible mass placement reduce twisting and demand.	Symmetry; avoid weak storeys; lighter upper levels

R4. Why is 'earthquake-resistant' more accurate than 'earthquake-proof'?

R5. Explain the difference between strength, stiffness and ductility. Why might maximising only stiffness be a poor strategy?

BEFORE TESTING

QuakeLab investigation routine

Use the same routine for every trial so your data can be compared.

Step	Action	Quality check
1	Build from Floor Joints. Join joints with Beams and/or Braces.	Is there a continuous path to the shake table?
2	Choose materials. Attach payload in 100 kg intervals.	Record material, mass and total cost.
3	Place up to four Sensors on joints or along members.	Record exactly where S1–S4 are located.
4	Set amplitude, frequency, duration and damping.	Change only the planned independent variable.
5	Start test. Observe motion and failures; inspect response graph.	Record maximum values with units.
6	Retest or modify after the test. Save evidence if required.	Label each version; do not overwrite observations.

Sensor meaning

A sensor reports signed horizontal movement (Δx) at its attachment point relative to the shake table. Positive and negative show direction. For comparison, use the maximum absolute value: the largest distance from zero.

Fair-test record

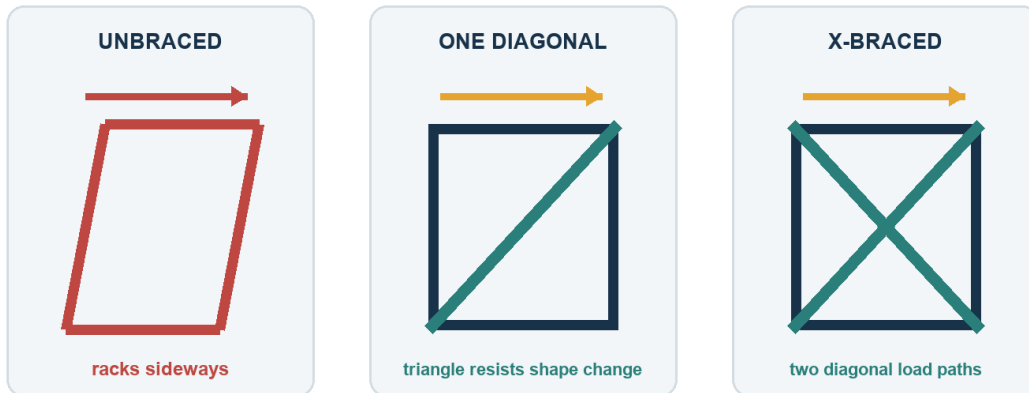
Independent variable	Dependent variable(s)	Controlled variables
what I change	what I measure	what I keep the same

Q0. Why should you reset or use the same starting structure before comparing trials?

INVESTIGATION 1

Frames and triangulation

How does diagonal bracing change lateral movement?



Controlled test

Build the same two-storey rectangular frame three times: A unbraced, B one diagonal per storey, C X-braced. Use Timber throughout, a 500 kg roof payload, amplitude 30 mm, frequency 1.50 Hz, duration 8 s, damping 8%. Place S1 on the roof and S2 on the first floor.

Prediction: Which design will have the lowest roof response? Why?

Design	Cost (\$)	Max S1 roof (mm)	Max S2 floor (mm)	Max drift (mm)	Failures	Survived?
A Unbraced						
B Diagonal						
C X-braced						

Q1. Make a claim about triangulation. Support it with two numerical results.

Q2. What trade-off appears when more bracing is added? Consider cost, material and access/openings.

INVESTIGATION 2

Materials: performance and cost

Does the most expensive material always give the best engineering solution?

Controlled test

Use your X-braced design from Investigation 1. Repeat the identical shake with every structural member changed to one material at a time. Keep geometry, sensors, 500 kg payload and shake settings unchanged.

Material	App cost / m	Strength	Stiffness	Weight	Damping
Timber	\$10	medium	medium	low	medium
Steel	\$30	high	high	high	low
Reinforced	\$22	high	high	medium	medium
Lightweight	\$7	low	low	low	medium-high

Material	Total cost (\$)	Max S1 (mm)	Max drift (mm)	Failures	Survived?
Timber					
Steel					
Reinforced					
Lightweight					

Q3. Which material gave the lowest movement? Which gave the best performance for cost? Use data.

Q4. Explain why material selection is a multi-criteria decision rather than a simple 'strongest wins' choice.

INVESTIGATION 3

Mass and where it is placed

How does payload position affect movement and structural demand?

Controlled test

Use one unchanged, two-storey X-braced design. Use Timber and the Investigation 1 shake settings. Compare 2000 kg at the first floor, 2000 kg at the roof, and 1000 kg at each level. Sensors remain at roof (S1) and first floor (S2).

Trial	Mass arrangement	Max S1 roof (mm)	Max S2 floor (mm)	Max drift (mm)	Failures	Survived?
A	2000 kg low					
B	2000 kg high					
C	1000 + 1000 kg					

Q5. Which mass arrangement produced the greatest roof response or drift? Quote measurements.

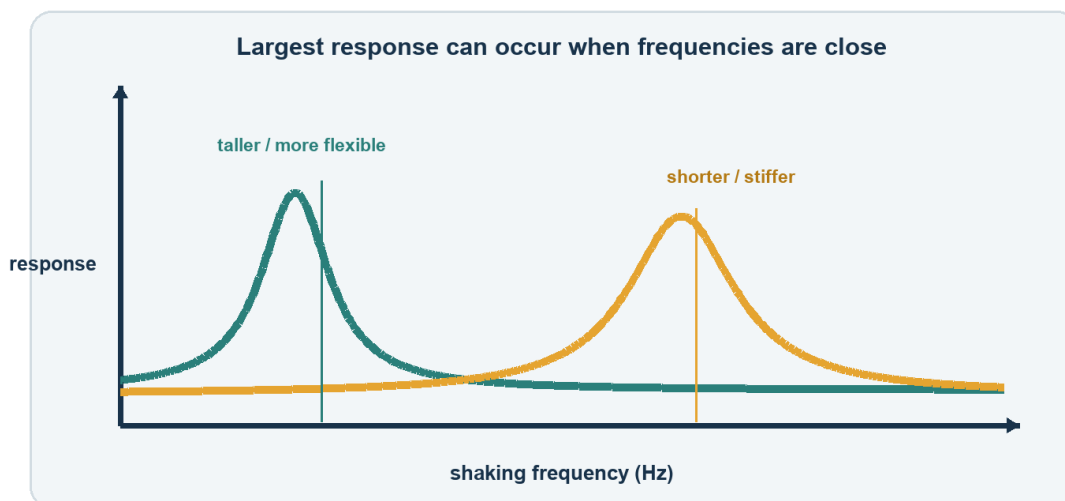
Q6. Use inertia and centre of mass to explain the pattern. Why can heavy upper levels be challenging?

Q7. Was total mass controlled in all three trials? Was mass position controlled? Identify the independent variable.

INVESTIGATION 4

Frequency, natural period and resonance

When can repeated shaking build a large response?



Every structure has natural modes of vibration. When forcing frequency is near a natural frequency, repeated pushes can add efficiently and response may grow. This is resonance. Height, stiffness, mass and damping all influence the response.

Frequency sweep

Use one unchanged design. Set amplitude 10 mm, duration 8 s and damping 8%. Test the listed frequencies. Record the app's calculated peak acceleration and the maximum absolute roof-sensor response.

Frequency (Hz)	0.50	1.00	1.50	2.00	3.00	4.00
Peak acceleration (g)						
Max S1 (mm)						

Q8. At which frequency was the response largest? Does this suggest a possible resonant region?

Q9. Why is constant amplitude not a perfectly fair resonance test? Hint: compare peak acceleration as frequency rises.

Q10. Predict how added damping would change the height and width of a resonance peak. Test one frequency if time permits.

OPEN-ENDED DESIGN TASK

Resilient Community Hub Challenge

Design a three-level community building that protects occupants while meeting constraints.

Scenario

Your town needs a compact community hub that may shelter people after an earthquake. The building needs useful floor space, must carry emergency supplies, and must be affordable. You will independently propose, test and justify a structural concept.

Non-negotiable constraints

Requirement	Constraint
Form	Three occupied levels, fixed to the shake table with Floor Joints
Payload	2000 kg total, placed in 100 kg intervals; justify its distribution
Materials	Use at least two app materials; total app cost $\leq$ \$750
Complexity	Maximum 24 structural members (beams + braces)
Measurement	Use four sensors: base/low, middle, upper and roof locations
Research	Use at least two earthquake-resistant principles and cite sources
Final test	60 mm amplitude • 2.00 Hz • 15 s • 8% damping
Performance	App survival + zero member failures + max roof $ \Delta x \leq 120$ mm

Engineering process

1 DEFINE → 2 IMAGINE → 3 PLAN → 4 BUILD → 5 TEST → 6 ANALYSE → 7 IMPROVE → 8 COMMUNICATE

Open-ended means

There is no single correct structure. A strong solution is one you can defend with measurements, structural reasoning, research, trade-offs and honest discussion of limitations.

CHALLENGE PLANNING

Concept sketch and design reasoning

Sketch clearly enough that your teacher or another student could reproduce your idea.

Front/elevation sketch — label floor joints, beams, braces, materials, masses and S1–S4	Key dimensions / notes

Decision	My choice	Reason / research link
Structural form and load path		
Bracing / lateral system		
Materials		
Mass distribution		
Sensor locations		

C1. Predict the part or storey most likely to experience the greatest movement or failure. Why?

CHALLENGE TESTING

Test, analyse and improve

Use a lower-risk diagnostic test before the final hazard test.

Diagnostic test

Suggested settings: 30 mm amplitude, 1.50 Hz, 8 s, 8% damping. Observe load paths and sensor differences. Change one major feature at a time so you can explain the effect.

Version	Change made	Cost (\$)	Max S1	Max S2	Max S3	Max S4	Drift	Failures
V1	baseline							
V2								
V3								

C2. Which change had the strongest evidence of improvement? Quote before-and-after data.

C3. Did any change improve one measure but worsen another? Explain the trade-off.

Final hazard test

Cost	Roof max $ \Delta x $	Max drift	Failed members	Height retained / app result	Challenge met?

Attach or paste graph/screenshot evidence here if your teacher requests it.

COMMUNICATE

Engineering evidence report

Answer in complete sentences. Use precise measurements and acknowledge limitations.

E1 — Claim. How well did your final design meet the brief? State a clear overall judgement.

E2 — Evidence. Give at least three relevant results, with units, including sensor response, drift/failure and cost.

E3 — Reasoning. Explain how two structural features caused the measured response. Use terms such as load path, stiffness, ductility, damping, inertia, triangulation or resonance correctly.

E4 — Research connection. How does one credible real-world source support or challenge a choice you made? Cite it.

E5 — Limitations. Identify two limitations of the simulator or your method. How could a real engineer obtain stronger evidence?

E6 — Next iteration. If given one more design cycle, what specific change would you test and what result would you expect?

ASSESSMENT

Student checklist and rubric

Use this before submitting your workbook and design evidence.

Criterion	High achievement	Sound achievement	Developing	Marks
Investigation method	Fair tests; variables and sensor positions clear; data complete with units	Mostly controlled tests; usable data	Variables unclear or data incomplete	/4
Structural understanding	Accurate, connected explanation of forces and ≥ 3 principles	Explains main principles with minor gaps	Mostly describes results without mechanism	/4
Data analysis	Patterns, anomalies and trade-offs supported by quantitative evidence	Uses relevant data to support conclusions	Limited or unsupported conclusions	/4
Research literacy	Two credible sources evaluated, paraphrased and cited; limitations recognised	Relevant sources cited and mostly explained	Sources weak, copied or not connected	/4
Design process	Constraints met; purposeful iterations; final decision strongly justified	Design tested and improved; most constraints met	Little iteration or weak justification	/4

TOTAL: ____ / 20

Submission checklist

- ☐ All four investigations include predictions, measurements and answers.
- ☐ My cited research is in my own words and URLs are recorded.
- ☐ My challenge design is labelled and meets the constraints, or exceptions are explained.
- ☐ I included evidence from more than one design version.
- ☐ My final report uses numerical evidence and discusses limitations.

My strongest design decision	My most useful next improvement

RESEARCH SOURCES

Trusted starting points

Use these to begin; your teacher may approve additional sources. Accessed 27 August 2026.

Geoscience Australia — [Earthquakes](#)

Australian earthquake processes, magnitude and intensity, local factors, impacts and case examples.

Geoscience Australia — [National Seismic Hazard Assessment](#)

How national earthquake-hazard information supports risk reduction and building-code decisions.

Geoscience Australia — [Resilience to the shake](#)

Australian work on the vulnerability and strengthening of older masonry buildings, including Adelaide context.

FEMA / NEHRP — [Earthquake-Resistant Design Concepts \(FEMA P-749\)](#)

Accessible technical explanations of strength, stiffness, ductility, bracing, isolation and energy dissipation.

U.S. Geological Survey — [What are the Effects of Earthquakes?](#)

How amplitude, frequency, duration, distance and local ground conditions influence shaking and damage.

U.S. Geological Survey — [How Seismic Waves Affect Different Size Buildings](#)

Why different building sizes respond differently to wave frequencies.

EarthScope Consortium (IRIS) — [Building resonance](#)

Natural period, forcing frequency and resonance explained through building response.

Source use

Do not copy sentences. Read, close the source, write the idea in your own words, reopen it to check accuracy, then record the organisation, title and URL.